Evidence Links of the Training Project — LavaRápido (Vehicle Washing)

Web and mobile booking and management platform for a car wash

Field	Value
Program / group	Technologist in Software Analysis and Development (ADSO), group 3145556
Center	Centro de la Industria, la Empresa y los Servicios — SENA, Huila Region
Training project	[To complete: official name of the training project]
Work team	Vehicle Washing Development Team ([To complete: confirm the official team name])
Members	Juan Camilo Guilombo Osorio, Brayan Estiven Patiño Cabrera, Merari Urbano Quintero
Instructor	Karol Daniela Correa
Delivery date	[To complete]
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1. Evidence Matrix

Documentation repositories used in the matrix

- **`lavarapido-docs`** (public, `main` branch): https://github.com/BrayanCABRERA123/lavarapido-docs
- **`vehicle-w-docs`** (private, `code-sena` organization, `docs` branch): https://github.com/code-sena/vehicle-w-docs/tree/docs. It is the source technical documentation of the project (SDD); the documents in `lavarapido-docs` are derived from it. Each evidence links the equivalent public section first, so there is always an accessible link.

Structure of `lavarapido-docs`: the general index (README) links every evidence and the change history (CHANGELOG) records each version. Each evidence has a numbered folder (`01_` to `11_`) and, inside it, `ES/` and `EN/`, each with `MD/` (Markdown and its `capturas/` screenshots), `PDF/`, and `WORD/`; the article (10) is bilingual in a single file, and this matrix (11) has Markdown and PDF. The matrix links the English Markdown version; the PDF and Word files are in the neighboring folders.

#	Evidence	Minimum verifiable content	Direct link	Access status	Remarks
1	Requirements specification report	Functional and non-functional requirements, user stories, acceptance criteria, traceability	SRS v7.1 (EN) · Functional requirements (SRS §5) · Non-functional requirements (SRS §6) · User stories per requirement (SRS §10) · Source in `vehicle-w-docs`: Functional requirements · Non-functional requirements · User stories	Public / Pending permissions (`vehicle-w-docs`)	The SRS follows IEEE 830 and ISO/IEC/IEEE 29148 and covers 41 functional requirements (RF-001 to RF-038) and use cases CU1 to CU30. Spanish version: `01_INFORME_DE_ESPECIFICACION_DE_REQUISITOS/ES/`.
2	Software analysis	Dynamic views, processes, use cases, sequence and state diagrams	Dynamic views analysis (EN) · Sequence, activity, and state diagrams · Source in `vehicle-w-docs`: UML diagrams (sources)	Public / Pending permissions (`vehicle-w-docs`)	The screenshots are in `02_DOCUMENTACION_DE_ANALISIS_DEL_SOFTWARE/EN/MD/capturas/`. The diagrams include the booking, payment, registration, reminder, assignment, and inspection sequences, and the booking and payment states.
3	Technical proposal of the software	Selected alternative, technologies, justification, scope, and constraints	Technical proposal (EN) · Design decisions (SRS §3.6) · Alternatives considered · Source in `vehicle-w-docs`: Architecture decisions (ADR-001 to ADR-015)	Public / Pending permissions (`vehicle-w-docs`)	The ADRs justify the microservices architecture, the database, the API Gateway, JWT, and the use of .NET for the payment service.

#	Evidence	Minimum verifiable content	Direct link	Access status	Remarks
4	Software design	Graphical interfaces, architecture, patterns, data model, and diagrams	Software design (EN) · Architectural model · Data model · API and system routes (technical manual §13) · Source in `vehicle-w-docs`: Architecture · Data model · OpenAPI contracts	Public / Pending permissions (`vehicle-w-docs`)	Hexagonal architecture pattern (ADR-007). Section 2.5 describes the responsive design. The screenshots are in `04_DOCUMENTACION_DE_DISENO_DEL_SOFTWARE/EN/MD/capturas/`.
5	Source code and software artifacts	Repositories of the web frontend, mobile app, backend, infrastructure, and database, with their branches	Code artifacts (EN) · see section 2	Public	The integration branch is `develop`. `main` is behind in every repository (see section 2).
6	Test documentation	Plan, test cases, results, execution evidence, defects, and fixes	Test documentation (EN) · E2E tests · Test plan · Source in `vehicle-w-docs`: Testing strategy	Public / Pending permissions (`vehicle-w-docs`)	Follows ISO/IEC/IEEE 29119-3. Automated (639) and end-to-end (19) tests run on 10/16/2026, functional cases on 10/09/2026, and the responsive design and mobile inspection check on 10/17/2026 (§3.2). The traceability matrix is in §5 and the findings in §6.
7	Implementation procedure	Reproducible installation, configuration, deployment, and verification	Implementation procedure (EN) · Azure deployment document (EN) · Support: Infrastructure README · Team guide	Public	Available in Markdown, PDF, and Word (ES and EN) in `07_PROCEDIMIENTO_DE_IMPLEMENTACION/`. The deployment document records the deployment of 10/16/2026 on Azure with its evidence screenshots.

#	Evidence	Minimum verifiable content	Direct link	Access status	Remarks
8	Corrective, preventive, and improvement actions	Findings, priority, owner, applied action, and closing evidence	Corrective, preventive, and improvement actions (EN) · Support: Preventive actions · Traceability between actions, risks, and technical debt · Source in `vehicle-w-docs`: Risks · Technical backlog	Public / Pending permissions (`vehicle-w-docs`)	Available in Markdown, PDF, and Word (ES and EN) in `08_ACCIONES_CORRECTIVAS_PREVENTIVAS_Y_DE_MEJORA/`.
9	Technical, installation, and user manuals	Prerequisites, configuration, use, and troubleshooting	Technical manual (EN) · Installation manual (EN) · User manual (EN) · Administrator manual (EN) · PDF of the manuals · Source in `vehicle-w-docs`: User manual	Public / Pending permissions (`vehicle-w-docs`)	Each manual is in Markdown, PDF, and Word (ES and EN), in the `09_MANUALES/` folder. Version 4.1 (10/17/2026), with real screenshots of the system; the `.env` secrets appear masked.
10	Project article	Problem, objectives, solution, methodology, results, and conclusions	APA 7 article (ES/EN) · APA 7 PDF · IEEE version · ACM version	Public	The three versions (APA 7, IEEE, and ACM) are in Markdown, PDF, and Word in the `10_ARTICULO_DEL_PROYECTO/` folder. Each file is bilingual (ES and EN in the same document), which is why this folder is not split by language.
11	Deployed application or demo	Access URL or demo environment and verification instructions	Application in production · Android app download · Deployment document (EN)	Public	Deployed on 10/16/2026 on an Azure virtual machine (Docker Compose, Caddy with Let's Encrypt HTTPS). The smoke tests and screenshots are in section 5 of the deployment document. The credentials are delivered through a private channel (see section 7).
12	Traceability and integration of evidence	Relationship between requirements, user stories, modules, tests, and versions	Traceability matrix (SRS §10) · Requirements–tests traceability (§5) · Source in `vehicle-w-docs`: Traceability matrix	Public / Pending permissions (`vehicle-w-docs`)	There are no release tags, so traceability to versions is done by branch (`develop`) and by commit.

2. Source Code Repositories

Working branch: **`develop`** in every code repository. There are no release tags. The "Last commit" and "`main` behind" columns were taken on 10/17/2026.

Component	Technology	Repository (`develop` branch)	Last commit	`main` behind	Access
Web frontend	Angular	Front-end-proyecto-web	`b3abfd9`	408 commits	Public
Mobile application	React Native (Expo)	LavadoVehicular-Movil	`41e2f29`	179 commits	Public
API Gateway	Spring Boot	lavarapido-api-gateway	`3c97264`	12 commits	Public
Security (authentication, JWT, roles)	Spring Boot	lavarapido-security-service	`a7867e3`	49 commits	Public
Customers and vehicles	Spring Boot	lavarapido-customer-service	`d0f4406`	6 commits	Public
Catalog and bookings	Spring Boot	lavarapido-booking-service	`affb46e`	45 commits	Public
Operations (operators and bays)	Spring Boot	lavarapido-operation-service	`c4d5cf9`	35 commits	Public
Payments and loyalty	.NET Core	lavarapido-payment-service	`694e1df`	39 commits	Public
Notifications	Spring Boot	lavarapido-notification-service	`23747c3`	34 commits	Public
Infrastructure (Docker, database, E2E tests)	Docker, SQL Server, RabbitMQ	lavarapido-infra	`95ce9ad`	54 commits	Public
Technical documentation (SDD)	Markdown, OpenAPI	vehicle-w-docs (`docs` branch)	`71f4a3f`	303 commits	Pending permissions
Documentary evidence	Markdown	lavarapido-docs (`main` branch)	last commit of `main`	—	Public

Database: shared SQL Server 2022, with one schema per service (ADR-003, ADR-009). Each Spring Boot service manages its migrations with Liquibase. The payment service migrations are in `lavarapido-payment-service/db`.

> Recommendation: before delivering, merge `develop` into `main` in each repository, or tell the evaluator to review `develop`, so that the evaluated version is consistent.

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3. Deployment and Technical Verification

Aspect	Value
Version (release tag)	`v1.0.0` pending. [To complete: create the tag in each repository when moving `develop` to `main`]
Environment	Academic production on Azure (Ubuntu 24.04 VM, Docker Compose with `docker-compose.prod.yml`), deployed on 10/16/2026. For development: local with Docker (`docker compose --profile app up -d --build` from `lavarapido-infra`). Details in the deployment document.
Public demo environment	https://lavarapidovehicular.northcentralus.cloudapp.azure.com

3.1 Role Matrix

Role	Main functions	Client where it is used
Administrator	Users and roles, catalog, promotions, schedules and bays, operators, bookings, inspection report, payments, reports	Web and mobile
Customer	Registration, vehicles, bookings, payments by QR or transfer, history, ratings, notifications	Web and mobile
Operator	Assigned services, schedule, execution, history, ratings received	Web and mobile

3.2 Health Endpoints (verified on 10/16/2026 in the local environment)

Service	Port	Endpoint	Result
API Gateway	8080	`/actuator/health`	`UP`
security-service	3001	`/actuator/health`	`UP`
customer-service	3002	`/actuator/health`	`UP`
booking-service	3003	`/actuator/health`	`UP`
operations-service	3004	`/actuator/health`	`UP`
payment-service	3005	`/health`	`ok` (checks SQL Server; answers 503 when the database does not respond)
notification-service	3006	`/actuator/health`	`UP`

3.3 Interactive API Documentation (Swagger/OpenAPI)

- **Spring Boot services (ports 3001–3004 and 3006):** `http://localhost:<port>/swagger-ui/index.html` answered `200` on 10/16/2026.
- **payment-service (.NET Core):** Swagger is only enabled in the `Development` environment. Under Docker it is not exposed (`404`). To see it, run the service with `ASPNETCORE_ENVIRONMENT=Development`.
- **Versioned contracts:** `07-api/contracts/openapi` in `vehicle-w-docs` (private). Public version: API and system routes, in the technical manual.

3.4 Smoke Tests

Test	Procedure	Result
Authentication	`POST http://localhost:8080/api/v1/auth/login` with a test user and check that it returns a JWT	200 with `accessToken` (10/16/2026, test account of the *dev* profile)
Read	With the JWT, query the service catalog or the customer's bookings	`GET /api/v1/catalog/services` and `GET /api/v1/bookings/me` answered 200 (10/16/2026)
Write	Create a booking and check that it appears in the history and that the notification arrives	Covered by `full-flow.test.js`: booking created, automatically assigned to the chosen operator, payment approved, service started and finished, rating, and every notice in the customer inbox (10/16/2026)
Automated E2E flow	`node --test "e2e/*.test.js"` in `lavarapido-infra` (with `E2E_OPERATOR_ID=2` in the local database)	19 of 19 tests passed in 7.3 s (10/16/2026)

3.5 Evidence Screenshots

Screenshot	Description	Link
Web screens (login, admin, customer, operator)	28 interface screenshots	SRS `capturas` folder
Dynamic views and modals	41 screenshots	Analysis `capturas` folder

Screenshot	Description	Link
Design (landing, login, booking, payments)	Screenshots of the final design	`capturas` folder
Real screenshots of the manuals	56 screenshots per language of the running system (10/16/2026): screens of the three roles, installation terminal, and data model diagram	Manuals `capturas` folder
Containers, logs, health checks, and migrations in production	Portainer, startup logs, `/actuator/health`, clean console, and Liquibase changesets (10/16/2026)	Deployment `capturas` folder

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4. Accessibility and Quality Checklist

- ☐ Every link opens correctly, without 404 errors or denied access, tested from an account outside the team.
- ☐ The instructor has read permission on `code-sena/vehicle-w-docs` (private) and checked it with their account.
- ☐ Each evidence in `lavarapido-docs` has its Markdown, PDF, and Word in ES and EN, and the PDF and Word files open correctly.
- ☐ The documents are readable: the Markdown renders on GitHub and the PDF files open without errors.
- ☐ Each repository identifies the project (README), the evaluated branch (`develop`), and the version (release tag).
- ☐ The documentation, repositories, and application are consistent: what the SRS and the manuals describe exists in the `develop` code.
- ☐ No simulations (mocks, `json-server`, sample data) are presented as real functionality.
- ☐ There are no credentials or sensitive data: `.env` is not versioned (only `.env.example`) and there are no tokens or passwords in the history.
- ☐ The exported PDF keeps the links active (click at least one per section).

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5. Acceptance Criteria

Criterion	Fulfillment condition	Result / remarks
Completeness	The 12 evidences have a direct and verifiable link	The 12 evidences have a link. Evidence 11 links the application deployed on Azure since 10/16/2026.
Accessibility	Every link opens for the instructor without asking for permissions	Partial: the links of `lavarapido-docs` and the code repositories are public (checked without logging in); those of `vehicle-w-docs` require giving the instructor read permission.
Traceability	Requirements → user stories → modules → tests → versions related	There is a matrix in §10 of the SRS, another in §5 of the test document, and the source matrix in `vehicle-w-docs`. Release tags are missing.
Consistency	The documentation matches the code and the running application	Local environment verified on 10/19/2026: every service `UP`, 655 automated tests, 19 end-to-end tests, and 42 functional cases passed, responsive design verified on 66 web screens and 15 app screens, and the screenshots of the manuals, SRS, analysis, and design taken again that day.
Security	No secrets or personal data exposed	Reviewed on 10/16/2026: no repository has `.env` files in its history and the versioned files only contain sample values (`.env.example`). The screenshots use test data.

Criterion	Fulfillment condition	Result / remarks
Presentation	Navigable document, readable tables, and PDF with active links	[To complete after exporting the PDF]

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6. Delivery

Format	Suggested name
Markdown	`11_ENLACES_DE_EVIDENCIAS/EN/MD/Evidence_Links_Project_VehicleWashing.md` (in Spanish: `ES/MD/Enlaces_Evidencias_Proyecto_VehicleWashing.md`)
PDF	`11_ENLACES_DE_EVIDENCIAS/EN/PDF/Evidence_Links_Project_VehicleWashing.pdf`; to deliver it, copy it with the date, for example `Evidence_Links_Project_VehicleWashing_2026-10-17.pdf`

The delivery channel (platform, email, shared folder, or other) is set by the instructor.

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7. Security Note

- Do not include in this document or in the repositories passwords, tokens, API keys, the content of `.env`, connection strings, or real personal data.
- **Test credentials:** if the evaluator needs to log in, the team will deliver through a private channel (direct message or institutional platform) one test user per role (administrator, customer, and operator). Those accounts are created only for the evaluation and are deactivated afterwards.
- Do not use links to local files (`C:\...`, `file://`), expired links, or links that require unresolved permissions.
- **Public repositories:** review the history (`.env` files, JWT secrets, database passwords) before publishing or delivering them.
- **Private repositories** (`vehicle-w-docs`): give the instructor read permission and check the access with their account.

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8. Reference Resources

- GitHub Docs: About repositories
- GitHub Docs: Setting repository visibility

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9. Reminder for the Team

> Update this matrix every time locations, branches, versions, or permissions change. After each change, export the PDF again and check the links again in the exported PDF.